Practice workbook · Small changes and percentage changes

Using dy is roughly (dy/dx) times dx to estimate the change, or percentage change, in a quantity from a small change in the variable.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-13

1. Question 5 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

5 y and x vary so that $y = xe^{-2x}$

Given that the value of x increases by 3%, use calculus to find, in terms of x, an estimate for the percentage change in y
Give your answer in the form $a(b - cx)$ where a, b and c are integers.

(6)

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1. Question 5 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

Question 5 continued

(Total for Question 5 is 6 marks)

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1. Question 5 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 5 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

5 A circle has radius $3r$ cm and area A cm^2

Given that the value of r increases by 0.05%

use calculus to find an estimate for the percentage increase in the value of A.

(5)

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2. Question 5 · part whole · 5 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 5 marks)

2. Question 5 · part whole · 5 marks · Working and answer

Working and answer

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3. Question 8 · part a · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

8 The volume of a sphere is $500\,cm^3$

(a) Calculate the radius, in cm to 3 significant figures, of the sphere.

(2)

The surface area of the sphere is increased by $20\,cm^2$

(b) Using calculus, find an estimate for the increase in the radius, in cm to 2 significant figures, of the sphere.

(5)

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3. Question 8 · part a · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

8 The volume of a sphere is $500\,cm^3$

(a) Calculate the radius, in cm to 3 significant figures, of the sphere.

(2)

The surface area of the sphere is increased by $20\,cm^2$

(b) Using calculus, find an estimate for the increase in the radius, in cm to 2 significant figures, of the sphere.

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4. Question 8 · part b · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.



Question 8 continued

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(Total for Question 8 is 7 marks)

4. Question 8 · part b · 5 marks · Working and answer

Working and answer

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5. Question 7 · part a · 7 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

7 The equation of a curve is $y = \sqrt{\frac{e^{4x}}{2x-3}}$

When x is increased to $(x + \delta x)$, y increases to $(y + \delta y)$ where δx and δy are small.

(a) Show that $\delta y \approx \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x$

(7)

Given that $x = 2.5$

(b) find an estimate, to 2 significant figures, of the value of δy when the value of x increases by 0.2%

(3)

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5. Question 7 · part a · 7 marks · Working and answer

Working and answer

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6. Question 7 · part b · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

7 The equation of a curve is $y = \sqrt{\dfrac{e^{4x}}{2x-3}}$

When x is increased to $(x + \delta x)$, y increases to $(y + \delta y)$ where δx and δy are small.

(a) Show that $\delta y \approx \dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\,\delta x$

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6. Question 7 · part b · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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6. Question 7 · part b · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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6. Question 7 · part b · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 10 marks)

6. Question 7 · part b · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

8 $y = \dfrac{2e^{3x+1}}{5x^2}$

(a) Find $\dfrac{dy}{dx}$

Give your answer in the form $\dfrac{Ae^{3x+1}(Bx-A)}{Cx^3}$ where A, B and C are prime numbers to be found.

(5)

The value of x increases by 2%

(b) Use your answer to part (a) to find an estimate, in terms of x, for the percentage change in y

Give your answer in the form $(Px-Q)$ where P and Q are integers.

(3)

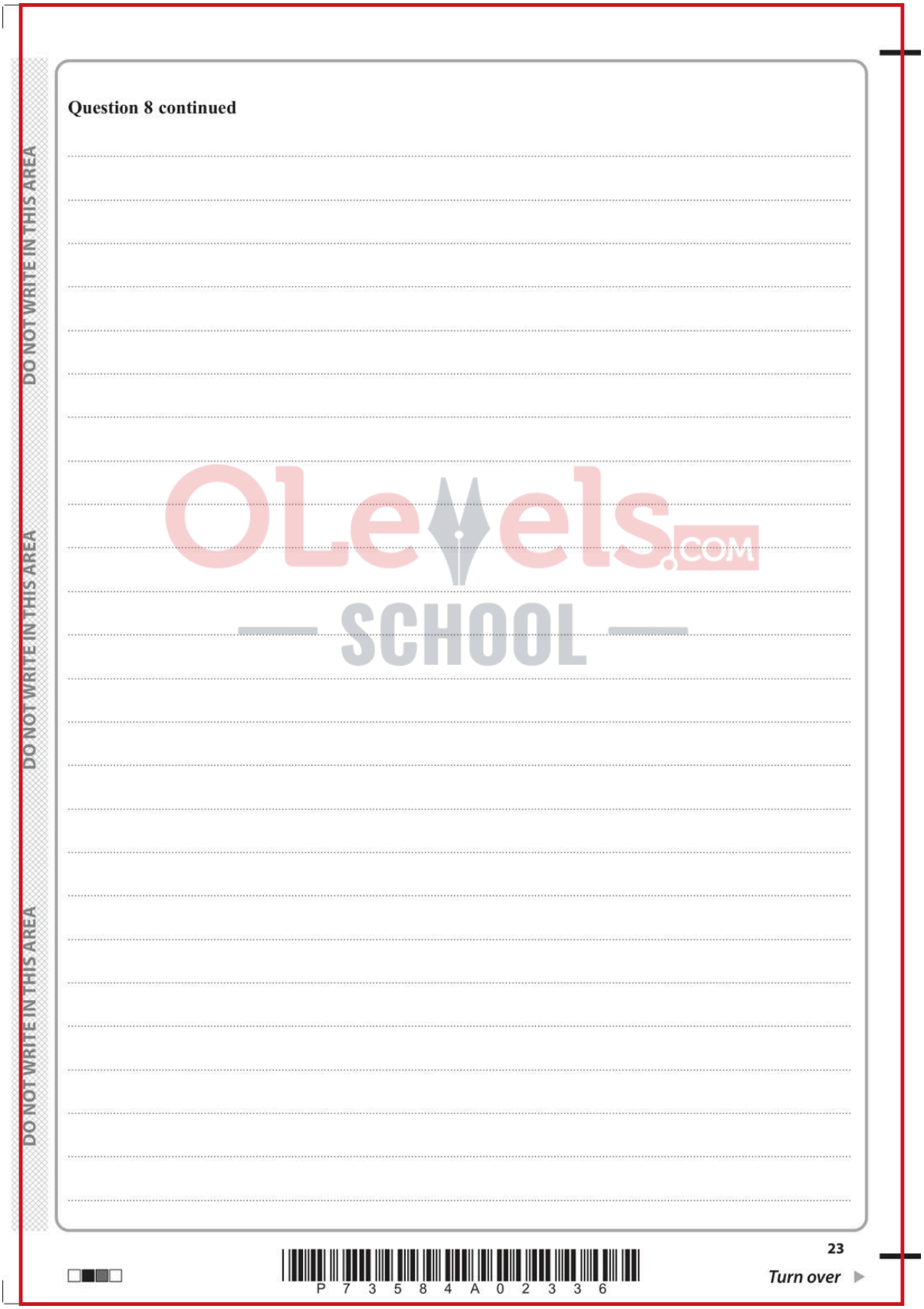
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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.



7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 8 continued

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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 8 marks)

7. Question 8 · part b · 3 marks · Working and answer

Working and answer

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